EXERCISE 2 AGGREGATE FUNCTIONS AND GROUPING

1. SELECT COUNT (*) AS total-employees
FROM employees;

total-employees
10

2. SUM() FUNCTION
SELECT sum (salary) AS total_IT_salary
FROM employees
WHERE department = 'IT';

total_IT_salary
220000

3. AVG() FUNCTION
SELECT AVG (salary) AS avg_hr_salary
FROM employees
WHERE department = 'HR';

avg_hr_salary
99000

4. MIN() and MAX FUNCTIONS
SELECT MIN (salary) AS lowest_salary
MAX (salary) AS highest_salary
FROM employees;

lowest_salary	highest_salary
48000	62000

5. GROUP BY STATEMENT

```
SELECT department, SUM (salary) AS total-salary
FROM employees
GROUP BY department;
```

department	total-salary
IT	220000
HR	99000
finance	119000
marketing	105000

6. GROUP BY and COUNT

```
SELECT city, COUNT (*) AS employee-count
FROM employees
GROUP BY city;
```

City	employee-count
New YORK	2
Chigago	3
Los Angeles	2
San Francisco	2
Houston	1

7. GROUP BY and ORDER BY

```
SELECT department, AVG (salary) AS avg-salary
FROM employees
GROUP BY department
ORDER BY avg-salary DESC;
```

department	avg-salary
IT	220000
finance	119000
marketing	105000
HR	99000

8. HAVING CLAUSE

```
SELECT department, SUM (salary) AS total_salary
FROM employees
GROUP BY department
HAVING SUM (salary) > 100000;
```

department	total_salary
IT	220000
finance	119000
marketing	105000

9. COMBINING GROUP BY, HAVING and ORDER BY

```
SELECT city, COUNT (*) AS employee_count
FROM employees
GROUP BY city
HAVING COUNT (*) > 1
ORDER BY employee_count DESC;
```

city	employee_count
Chicago	3
New York	2
Los Angeles	2
San Francisco	2

10. COMBINING AGGREGATE FUNCTIONS

```
SELECT department, AVG (salary) AS avg_salary
FROM employees
GROUP BY department
ORDER BY avg_salary DESC
LIMIT 1;
```

department	avg_salary
IT	220000